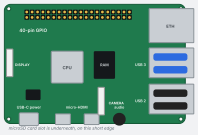
RAQIB · AI INSPECTION ROBOT · BUDGET BUILD

Parts & Wiring Atlas

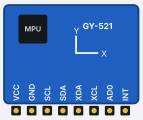
Every sensor and board in colour with its name and real size, the Raspberry Pi pin by pin, the robot's structure layer by layer, and how each part is connected.



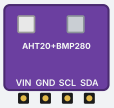
Raspberry Pi 4



HC-SR04



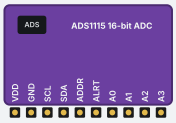
MPU6050



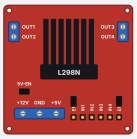
AHT20 + BMP280



MQ-135



ADS1115



L298N



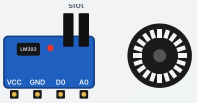
XL4015



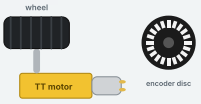
USB webcam



PAM8403



Encoder



TT motor

Ahmed Mahram Saeed Saeed Hadi
Final-year IoT project · UNISIRAJ

Atlas 1.0 · October 2026
Matches the Raqib software 2.4 and the build guide

Inside this atlas

1. **Every part** - 23 parts in six colour groups, each with a big picture, its name, how to recognise it and where its pins go
2. **The boards** - the Raspberry Pi 4 and its ports, all 40 pins, and close-ups of the L298N, XL4015, ADS1115, HC-SR04 and the hub board
3. **The robot's structure** - exploded view, side, top, bottom and front views
4. **How everything connects** - the system map, one wiring diagram per group, the complete harness and a wire-by-wire table

About the pictures: they are drawings of the common versions of each board, in their real colours and layout. Your board may be a slightly different colour or have its pins in another order: **always read the labels printed on the board**.

Wire colours used in every diagram

5 V · 3.3 V · GND · SDA · SCL · battery + · motor · PWM speed · signal · audio

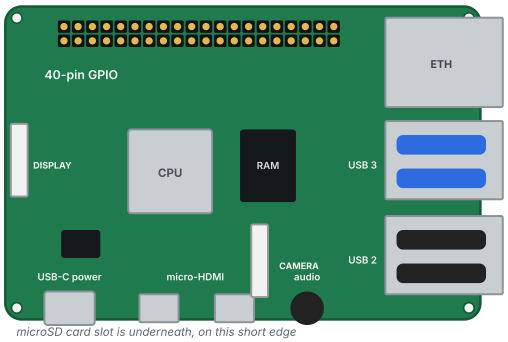
Use the same colours for your real wires where you can: mistakes become easy to spot.

Colour groups: The brain · Power · Moving · Sensing · Camera and sound · Wiring and building

The brain

The computer and its memory card.

1 Raspberry Pi 4 Model B (2 GB)



microSD card slot is underneath, on this short edge

WHAT IT DOES

The robot's computer. Runs the robot program, its own server for the phone app, and talks to the AI.

HOW TO RECOGNISE IT

Green board the size of a credit card. Silver USB and Ethernet ports on one short edge, 40 gold pins along one long edge.

REAL SIZE

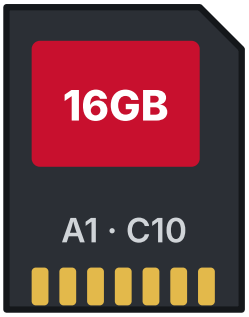
85 × 56 mm

CONNECTIONS

40-pin header	all sensors and the L298N (see the pin map)
USB-C	5.1 V from the XL4015 (cut USB cable)
USB 3 (blue)	USB webcam
3.5 mm jack	PAM8403 amplifier

💡 The microSD card slot is underneath, on the short edge without ports.

2 microSD card 16 GB



WHAT IT DOES

Holds Raspberry Pi OS and all the robot software.

HOW TO RECOGNISE IT

Fingernail-size card; buy 'A1' or 'Class 10'.

REAL SIZE

15 × 11 mm

CONNECTIONS

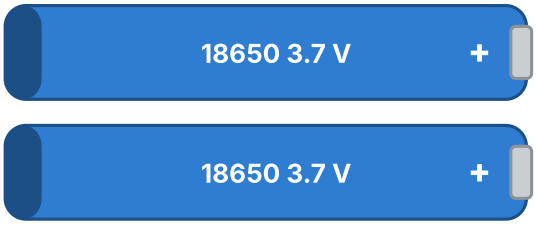
card	slot under the Pi

💡 Write it with Raspberry Pi Imager on your PC.

Power

One battery pack powers everything: motors directly, the Pi through a 5 V converter.

3 18650 Li-ion cells (×2)



WHAT IT DOES
Store the energy for the whole robot.

HOW TO RECOGNISE IT
Cylinder like a fat AA battery, usually blue, green or purple wrapping, flat or button top.

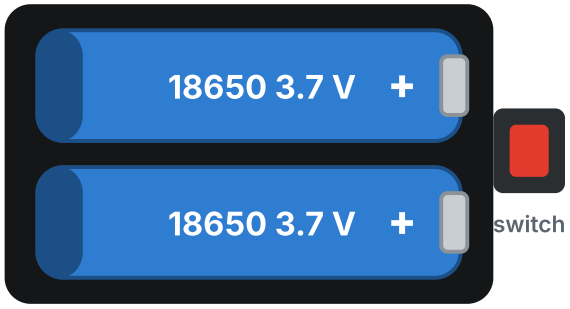
REAL SIZE
18 mm × 65 mm each, 3.7 V

CONNECTIONS

● **+ (button end)**	+ mark in the holder
● **– (flat end)**	spring in the holder

💡 Real cells are 2500-3500 mAh. '9900 mAh' cells are fake and unsafe.

4 2 × 18650 holder with switch



WHAT IT DOES
Joins the two cells into a 7.4 V pack; its switch turns the robot on and off.

HOW TO RECOGNISE IT
Black plastic box with two slots, a small slide switch and a red + black lead.

REAL SIZE
about 77 × 41 mm

CONNECTIONS

● **red lead +**	XL4015 IN+, L298N +12V, battery divider
● **black lead –**	XL4015 IN−, L298N GND

💡 Full = 8.4 V, empty ≈ 6.6 V.

5 XL4015 step-down converter



WHAT IT DOES
Turns the battery's 7-8 V into a steady 5.1 V for the Pi.

HOW TO RECOGNISE IT
Blue board with a big black coil, a square XL4015 chip and a small blue box with a brass screw (trimmer).

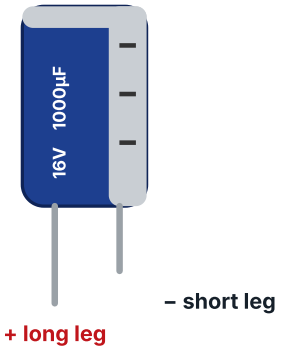
REAL SIZE
about 54 × 23 mm

CONNECTIONS

IN+ / IN−	battery + / − (1000 µF across them)
OUT+ / OUT−	red / black of the cut USB cable → Pi

💡 Turn the brass screw until OUT reads 5.1 V BEFORE the Pi is plugged in.

6 1000 µF capacitor



WHAT IT DOES
A small energy tank: stops motor starts from dipping the Pi's power.

HOW TO RECOGNISE IT
Blue or black cylinder with a white stripe and '−' marks on one side.

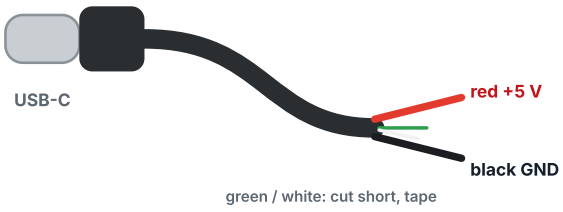
REAL SIZE
about 10 × 20 mm

CONNECTIONS

long leg (+)	XL4015 IN+
short leg, stripe (−)	XL4015 IN−

💡 Polarity matters: the stripe side goes to minus.

7 USB-A → USB-C cable (A end cut off)



WHAT IT DOES
Carries the 5.1 V from the XL4015 into the Pi's power socket.

HOW TO RECOGNISE IT
Any 3 A phone cable. Inside: red, black, green and white wires.

REAL SIZE
20-30 cm

CONNECTIONS

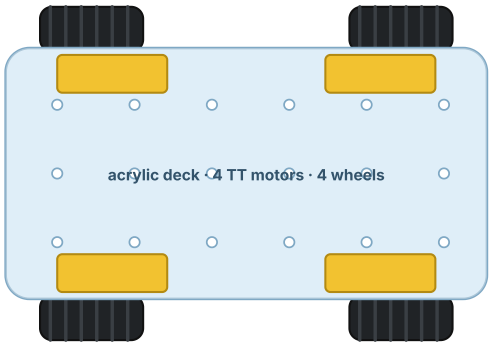
red	XL4015 OUT+
black	XL4015 OUT−
USB-C plug	Pi USB-C power

💡 Green and white are data wires: cut them short and tape them.

Moving

Body, wheels, motor driver and the wheel-turn counters.

8 4WD chassis kit



WHAT IT DOES

The robot's body: two acrylic decks, four motors with wheels, brass standoffs.

HOW TO RECOGNISE IT

Two clear or coloured acrylic plates with many holes, 4 yellow TT motors, 4 rubber wheels, brass posts and screws.

REAL SIZE

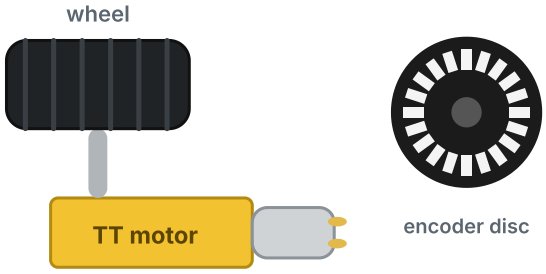
about 25 × 15 cm

CONNECTIONS

● **4 motors**	under the bottom deck, 2 per side
● **standoffs**	bottom deck → top deck

💡 Four driven wheels steer like a tank: left pair and right pair at different speeds.

9 TT gear motors (×4) + encoder discs



WHAT IT DOES

Each turns one wheel. The 20-slot disc on the two front motors lets the robot count how far it went.

HOW TO RECOGNISE IT

Yellow plastic gearbox with a small metal motor; shaft out of both sides.

REAL SIZE

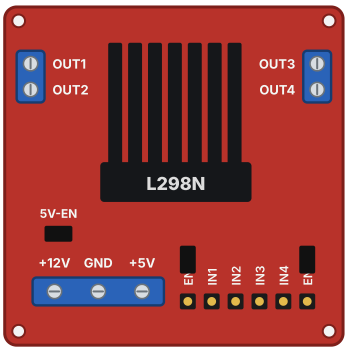
70 × 22 × 18 mm

CONNECTIONS

● **left pair red / black**	L298N OUT1 / OUT2 (front + rear joined)
● **right pair red / black**	L298N OUT3 / OUT4

💡 Join the two motors of a side red-to-red, black-to-black; a side that runs backwards: swap its two wires.

10 L298N motor driver



WHAT IT DOES
Switches battery power to the motors forwards or backwards, at the speed the Pi asks.

HOW TO RECOGNISE IT
Red board with a big black finned heatsink, blue screw terminals on three sides, 6 pins on the front.

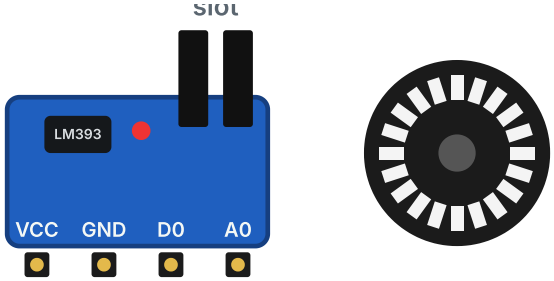
REAL SIZE
43 × 43 mm

CONNECTIONS

+12V / GND	battery + / –
GND	also to Pi pin 9
OUT1/OUT2	left motor pair
OUT3/OUT4	right motor
ENA / ENB	Pi pins 33 / 35 (speed)
IN1-IN4	Pi pins 11, 13, 15, 16 (direction)

💡 Pull off the ENA and ENB jumpers. Keep the 5V-EN jumper.

11 Speed encoder module (×2)



WHAT IT DOES
Gives one pulse each time a slot of the disc passes: the wheel's odometer.

HOW TO RECOGNISE IT
Small blue board with a black U-shaped slot sensor and an LM393 chip.

REAL SIZE
32 × 14 mm

CONNECTIONS

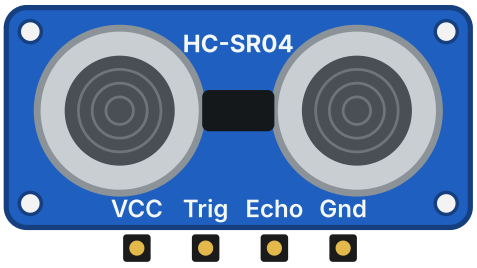
VCC	hub 3V3
GND	hub GND
D0	Pi pin 29 (left) / 31 (right)
A0	not used

💡 The disc must spin inside the slot without touching it.

Sensing

Distance, turning, climate and air-quality sensors.

12 HC-SR04 ultrasonic sensor (×3)



WHAT IT DOES

Measures distance with sound (2 cm-4 m): stops the robot before obstacles.

HOW TO RECOGNISE IT

Blue board with two silver 'eyes' (speaker and microphone), 4 pins at the bottom.

REAL SIZE

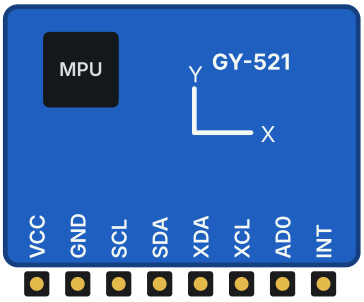
45 × 20 mm

CONNECTIONS

VCC	hub 5V
Trig	Pi pin 32 / 18 / 38 (left / centre / right)
Echo	1k/2k divider → Pi pin 36 / 22 / 40
Gnd	hub GND

💡 Echo is 5 V: never connect it to the Pi without the divider.

13 MPU6050 motion sensor (GY-521)



WHAT IT DOES

Gyroscope + accelerometer: knows how far the robot has turned.

HOW TO RECOGNISE IT

Tiny blue board with a square black chip and 8 pins in a row.

REAL SIZE

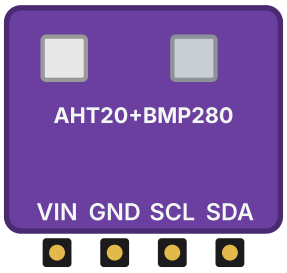
21 × 16 mm

CONNECTIONS

VCC	hub 3V3
GND	hub GND
SCL	hub SCL (Pi pin 5)
SDA	hub SDA (Pi pin 3)

💡 Mount it flat near the middle; I²C address 0x68.

14 AHT20 + BMP280 climate sensor



WHAT IT DOES
Temperature, humidity (AHT20) and air pressure (BMP280) on one board.

HOW TO RECOGNISE IT
Small purple or blue board with a white square chip (AHT20) and a tiny silver one (BMP280).

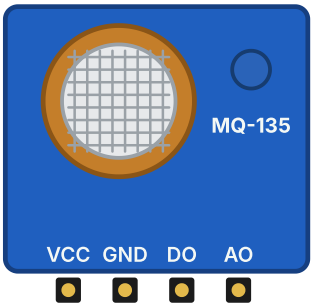
REAL SIZE
about 15 × 12 mm

CONNECTIONS

VIN	hub 3V3
GND	hub GND
SCL	hub SCL
SDA	hub SDA

💡 Two I²C addresses: 0x38 and 0x77 (or 0x76). Keep it away from warm parts.

15 MQ-135 gas sensor



WHAT IT DOES
Heated sensor whose voltage rises with CO_2 and gases: the air-quality check.

HOW TO RECOGNISE IT
Blue board with a silver mesh 'can' on top, a red LED and 4 pins.

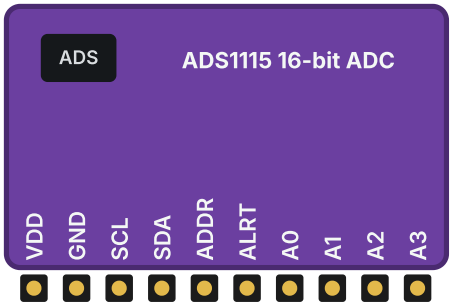
REAL SIZE
32 × 22 mm

CONNECTIONS

VCC	hub 5V (heater)
GND	hub GND
AO	10k/20k divider → ADS1115 A0
DO	not used

💡 It gets hot. First use: leave it on 24-48 h.

16 ADS1115 16-bit ADC



WHAT IT DOES
Turns voltages into numbers for the Pi: the gas sensor and the battery level.

HOW TO RECOGNISE IT
Small purple or blue board with 10 pins along one edge.

REAL SIZE
28 × 17 mm

CONNECTIONS

VDD	hub 3V3
GND	hub GND
SCL / SDA	hub SCL / SDA
ADDR	GND (address 0x48)
A0	MQ-135 divider
A1	battery divider (10k/4.7k)

💡 Power it from 3.3 V so it is safe on the Pi's I²C bus.

Camera and sound

Seeing, hearing and speaking.

17 USB webcam with microphone



WHAT IT DOES
The robot's eyes and ears: photos, live video and sound.

HOW TO RECOGNISE IT
Small black camera on a clip, one USB cable, tiny mic holes on the front.

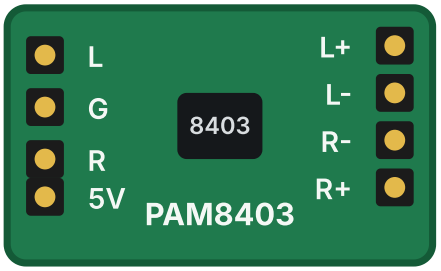
REAL SIZE
about 70-90 mm wide

CONNECTIONS

USB plug	Pi USB 3 port (blue)

💡 Choose 'UVC / plug and play' with a built-in mic. It is fixed: the robot turns to look.

18 PAM8403 amplifier



WHAT IT DOES
Makes the Pi's audio loud enough for the speaker.

HOW TO RECOGNISE IT
Tiny green board (thumbnail size) with a small chip and pins or pads.

REAL SIZE
about 21 × 18 mm

CONNECTIONS

5V / GND	hub 5V / GND
L / G	3.5 mm cable: tip / shield
L+ / L−	speaker

💡 Use only the left channel.

19 Speaker 3 W 4 Ω



3 W 4 Ω speaker

WHAT IT DOES
The robot's voice.

HOW TO RECOGNISE IT
Round, about 40 mm, black cone, two solder tabs.

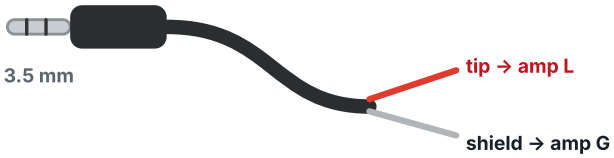
REAL SIZE
40 mm

CONNECTIONS

+ tab	PAM8403 L+
– tab	PAM8403 L–

💡 Face it outwards at the back of the top deck.

20 3.5 mm audio cable (one end cut)



WHAT IT DOES
Takes the sound from the Pi's audio jack to the amplifier.

HOW TO RECOGNISE IT
Ordinary headphone-plug cable.

REAL SIZE
20-30 cm

CONNECTIONS

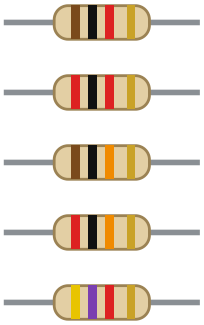
plug	Pi 3.5 mm audio jack
tip wire	PAM8403 L
shield	PAM8403 G

💡 If you are unsure which wire is the tip, test with the multimeter (continuity).

Wiring and building

What holds the robot and its wires together.

21 Resistors



WHAT IT DOES
Voltage dividers: protect the Pi from 5 V and measure the battery.

HOW TO RECOGNISE IT
Small beige or blue tubes with coloured bands.

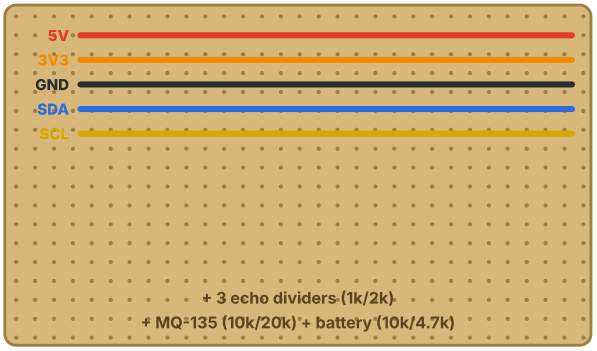
REAL SIZE
about 6 mm long

CONNECTIONS

1 kΩ ×3 + 2 kΩ ×3	the three Echo dividers
10 kΩ + 20 kΩ	MQ-135 divider
10 kΩ + 4.7 kΩ	battery divider

💡 No 2 kΩ? Two 1 kΩ in series. Read the colour bands (picture).

22 Hub board (you make it)



WHAT IT DOES
One place where 5 V, 3.3 V, GND, SDA and SCL meet, plus the 5 dividers.

HOW TO RECOGNISE IT
Brown perfboard with five soldered rails and header strips.

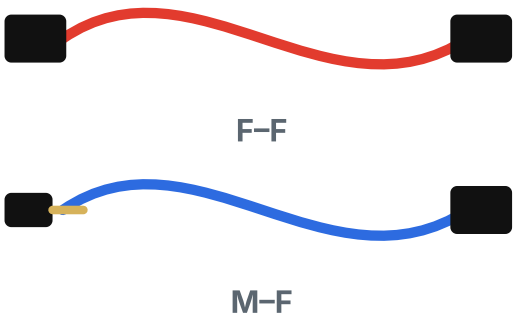
REAL SIZE
about 7 × 9 cm

CONNECTIONS

5V / 3V3 / GND	Pi pins 2 / 1 / 6
SDA / SCL	Pi pins 3 / 5

💡 Label each rail with a marker.

23 Jumper wires



WHAT IT DOES
Plug-in connections: no soldering for most parts.

HOW TO RECOGNISE IT
Rainbow ribbon of wires; F = socket end, M = pin end.

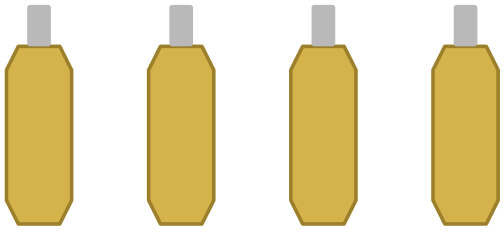
REAL SIZE
10-20 cm

CONNECTIONS

F-F	pin to pin (Pi to sensor)
M-F	pin to socket

💡 Use red for 5 V, orange for 3.3 V, black for GND.

24 Standoffs and screws



M3 standoffs 35-40 mm + screws

WHAT IT DOES
Hold the top deck above the bottom deck, and the boards on the decks.

HOW TO RECOGNISE IT
Brass or nylon hexagon posts with threads.

REAL SIZE
M3, 35-40 mm long

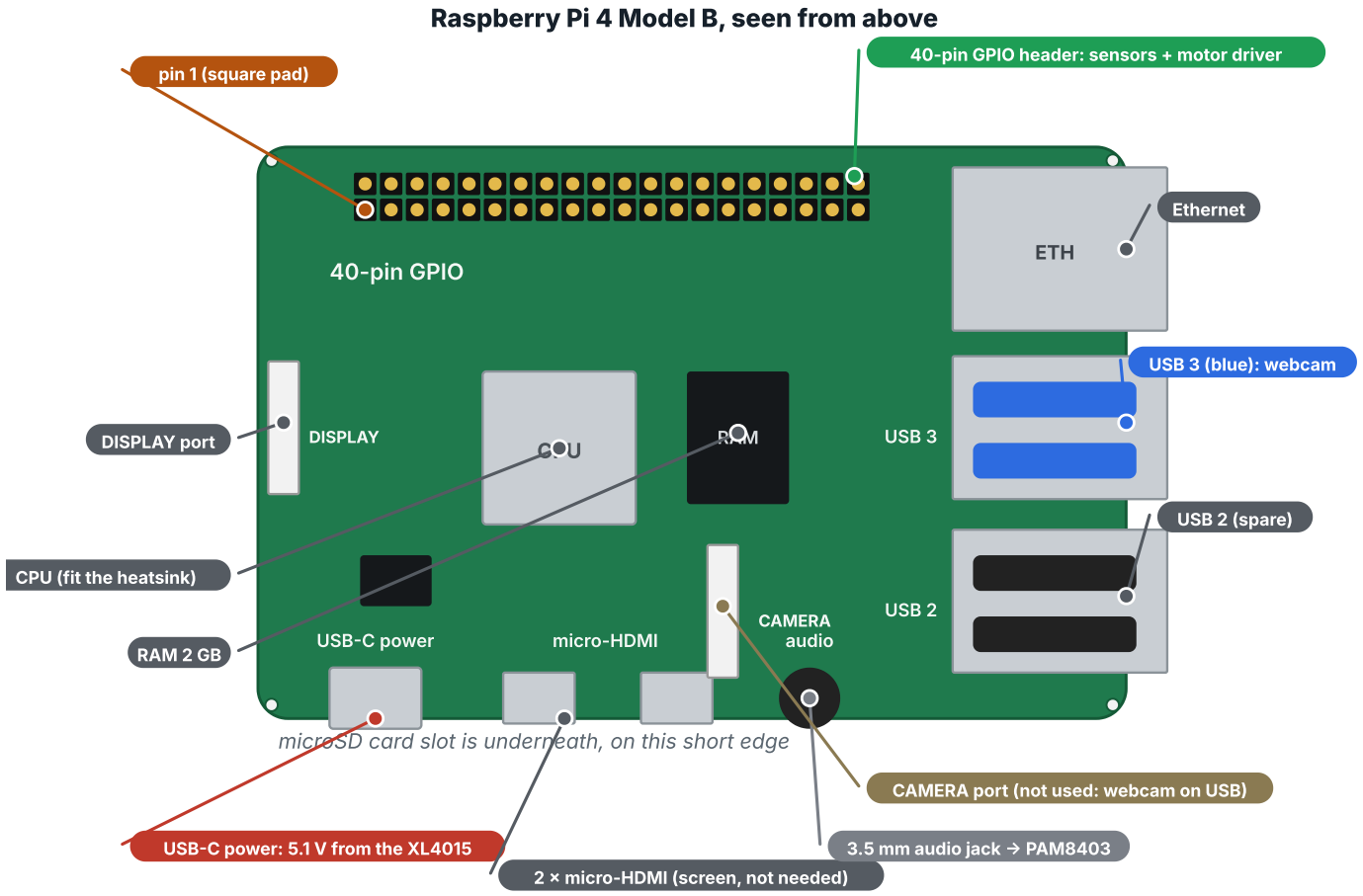
CONNECTIONS

M3 posts	between the decks
M2.5 screws	Raspberry Pi

💡 Nylon washers under boards stop short circuits.

B The boards

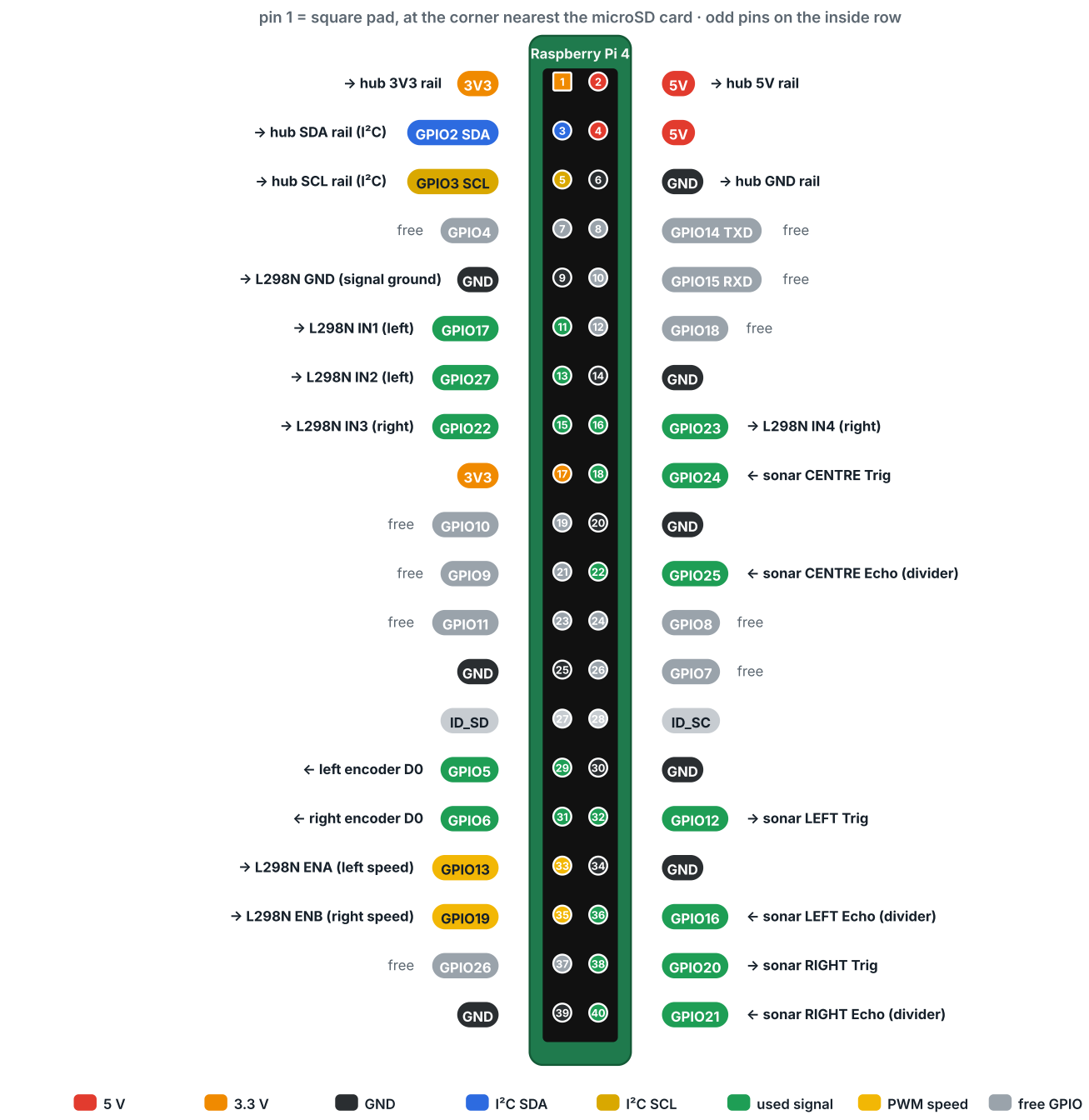
The Raspberry Pi 4 is the centre of everything: every other part connects to its pins or ports.



The Raspberry Pi 4 and what each port is used for in this robot.

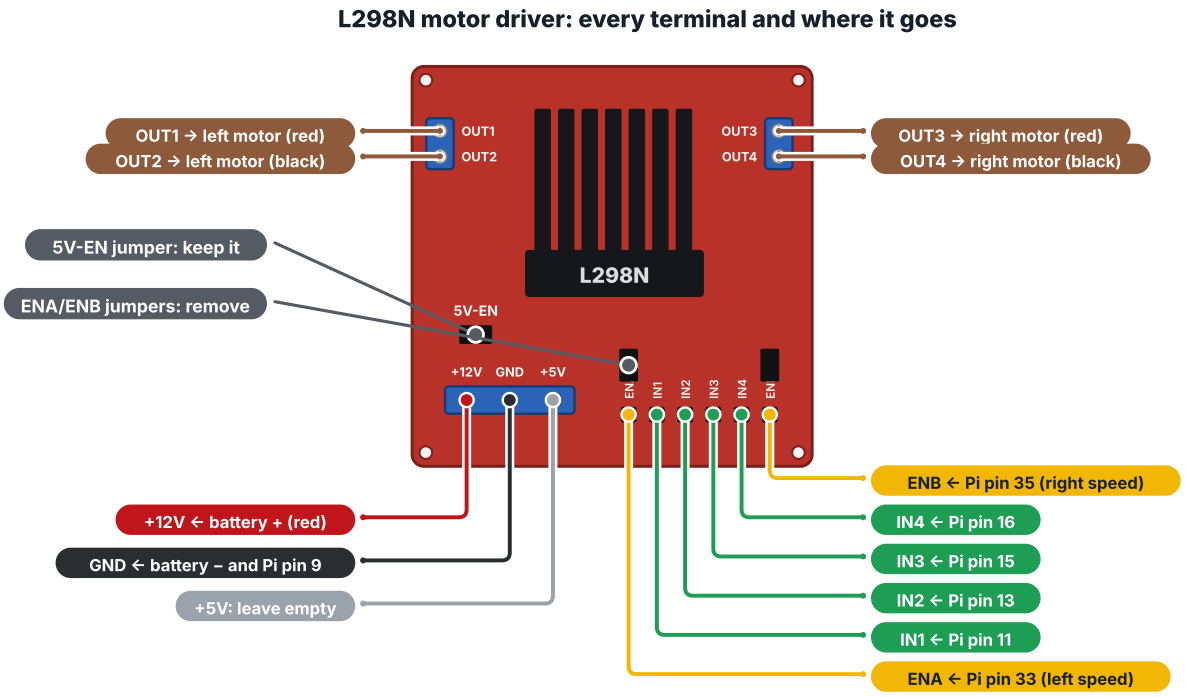
All 40 pins of the Raspberry Pi header

Pin numbers are counted on the board (1-40). Coloured pins are used by the robot; grey ones are free.



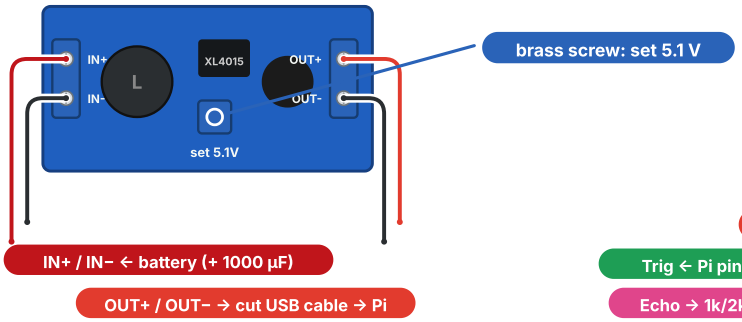
The 40-pin header, top view, USB ports pointing down the page.

Board close-ups

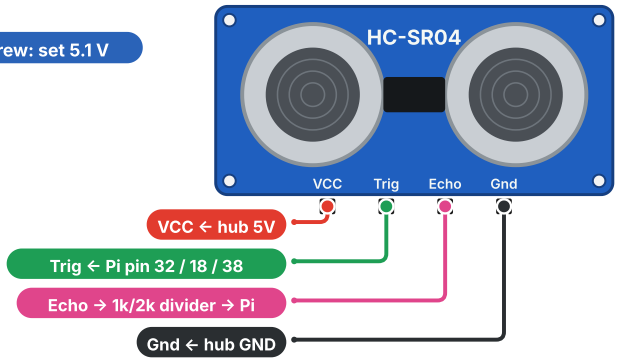


The L298N: battery and motors on the screw terminals, the Pi on the 6 pins.

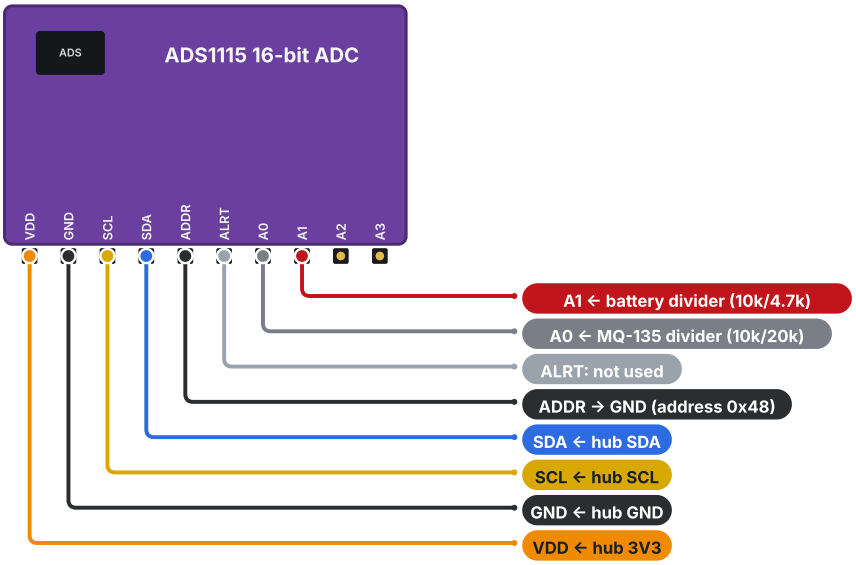
XL4015 5 V converter



HC-SR04 ultrasonic sensor

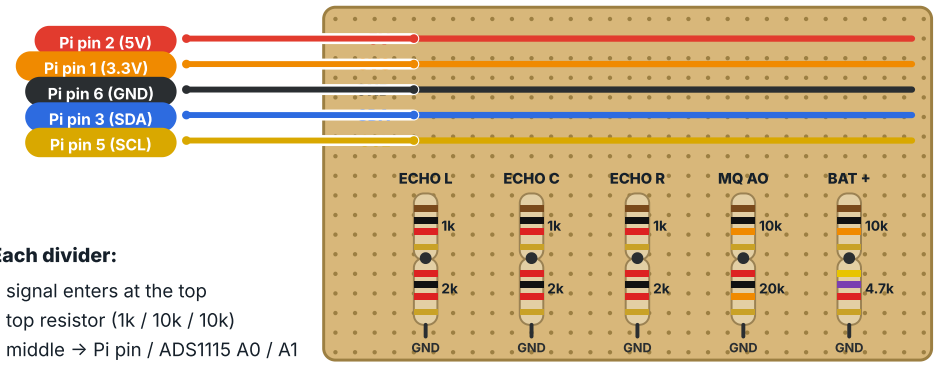


ADS1115 analog-to-digital converter



XL4015, HC-SR04 and ADS1115: every pin and where it goes.

The hub board (you make it)



Each divider:

- signal enters at the top
- top resistor (1k / 10k / 10k)
- middle → Pi pin / ADS1115 A0 / A1
- bottom (2k / 20k / 4.7k) → GND
- 5 V → 3.3 V · battery 8.4 V → 2.7 V

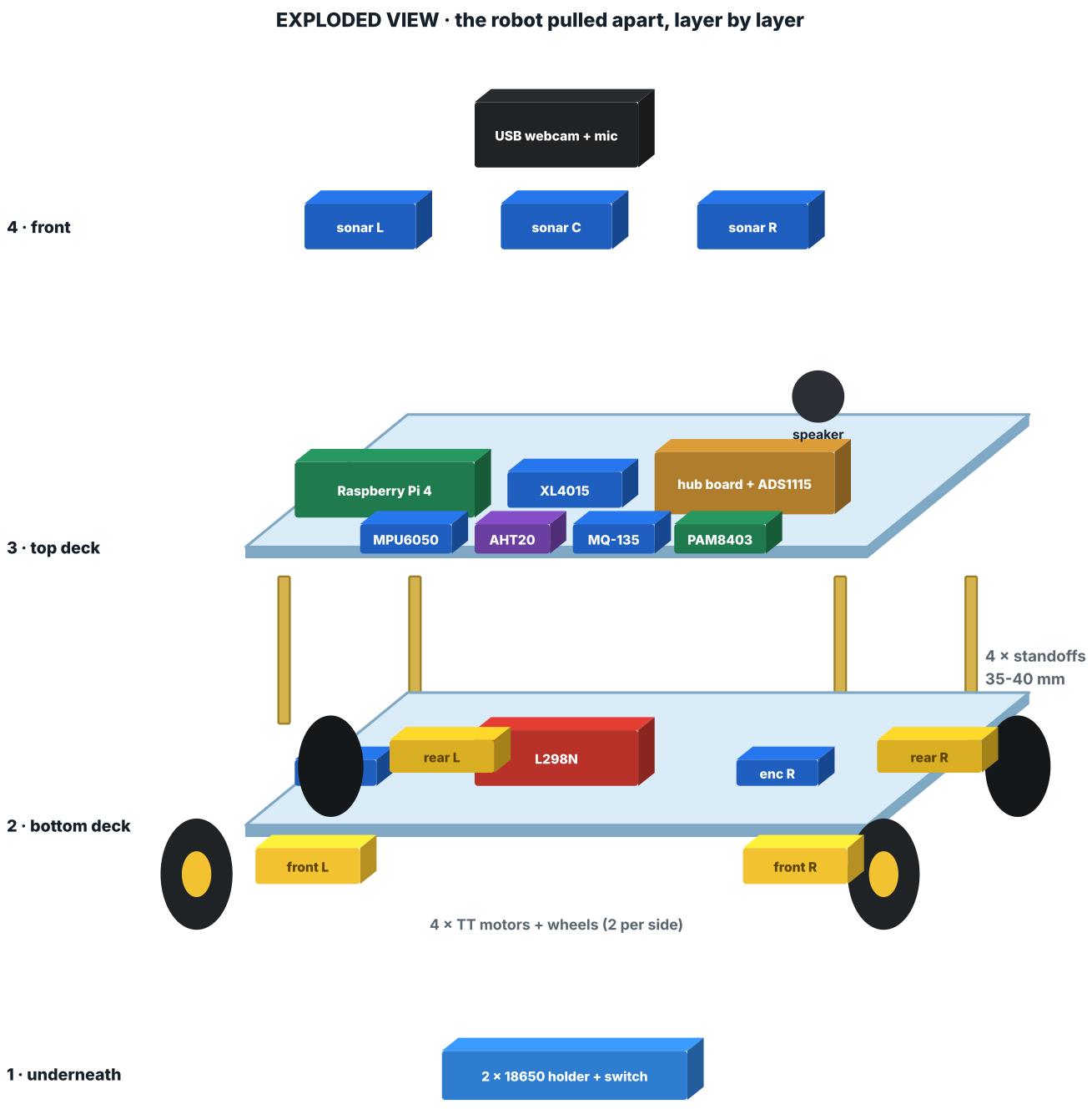
Solder the rails as bare wire along the strips;
every module plugs into the rails with jumper wires.

Five rails on top, the five voltage dividers below.

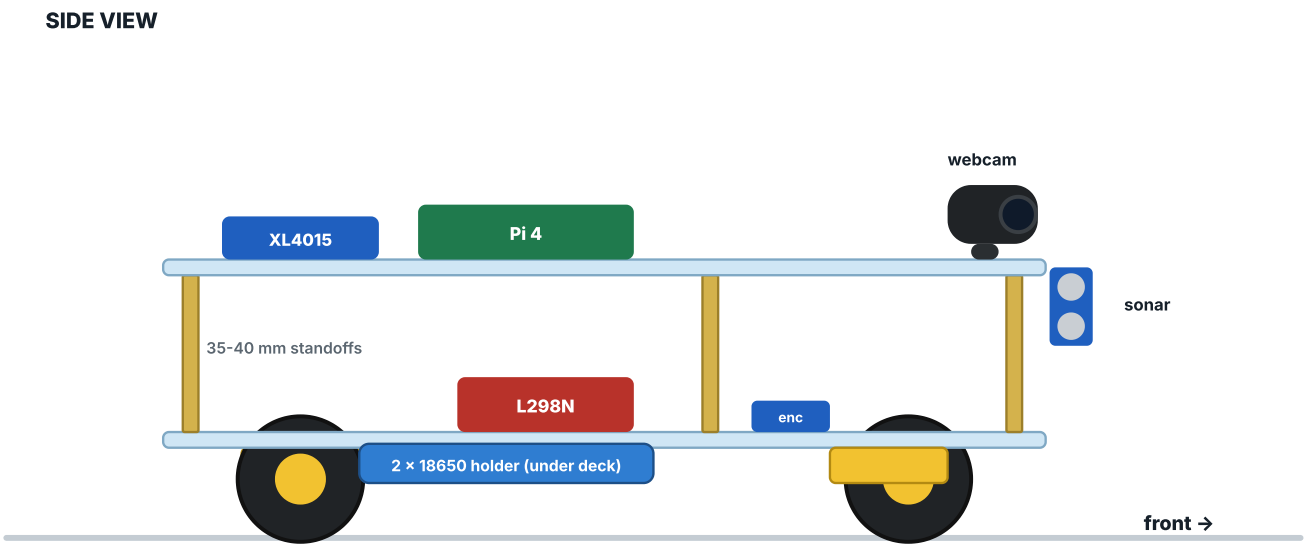
The hub board saves wires: the Pi gives 5 V, 3.3 V, GND, SDA and SCL to the hub once, and every sensor plugs into the hub. The dividers turn the 5 V sonar Echo and MQ-135 signals into safe 3.3 V, and the battery's 8.4 V into 2.7 V.

C The robot's structure

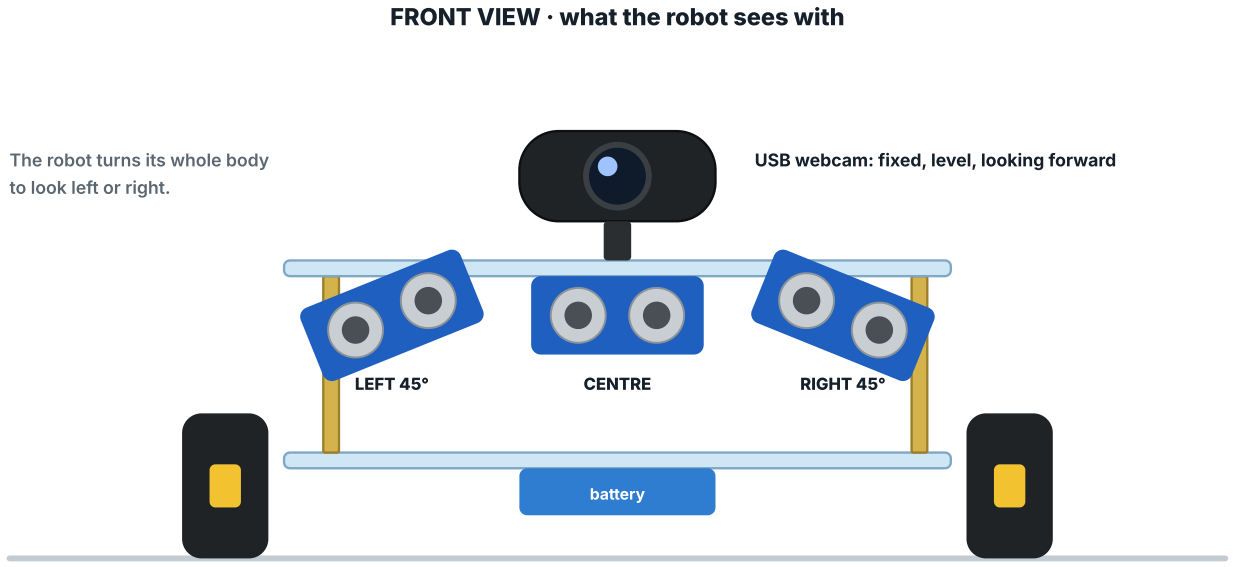
Two acrylic decks on standoffs. The battery hangs underneath, the motors and motor driver sit on the bottom deck, the computer and sensors on the top deck, and the distance sensors and webcam face forward.



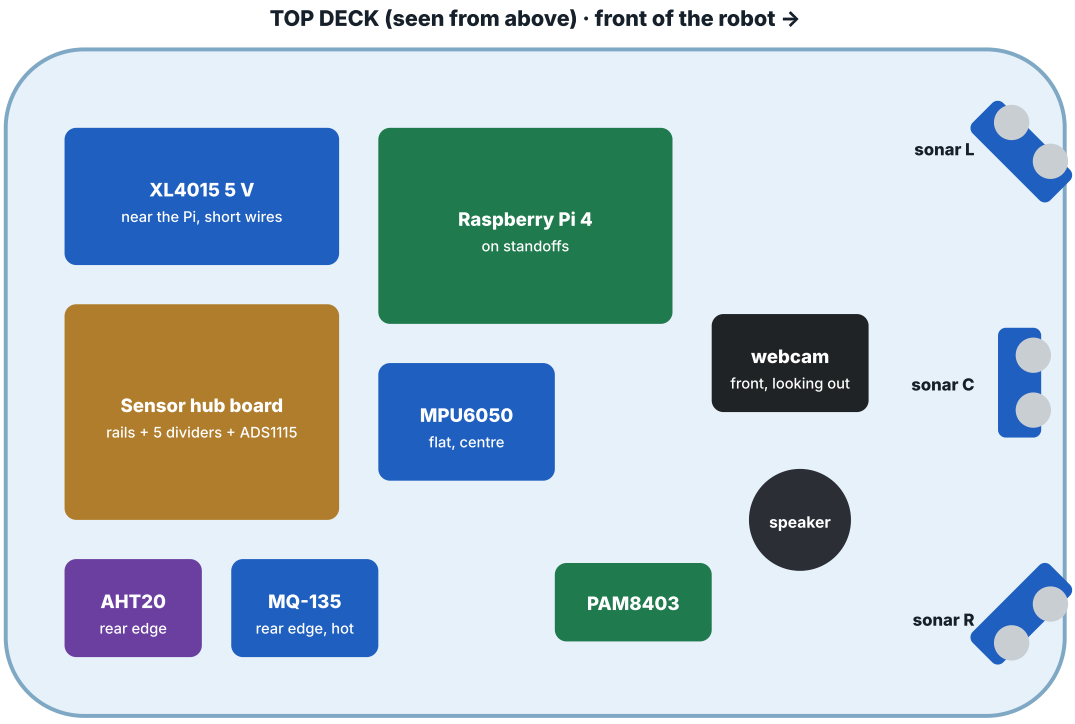
Exploded view: 1 battery, 2 bottom deck, 3 top deck, 4 front sensors.



Side view.

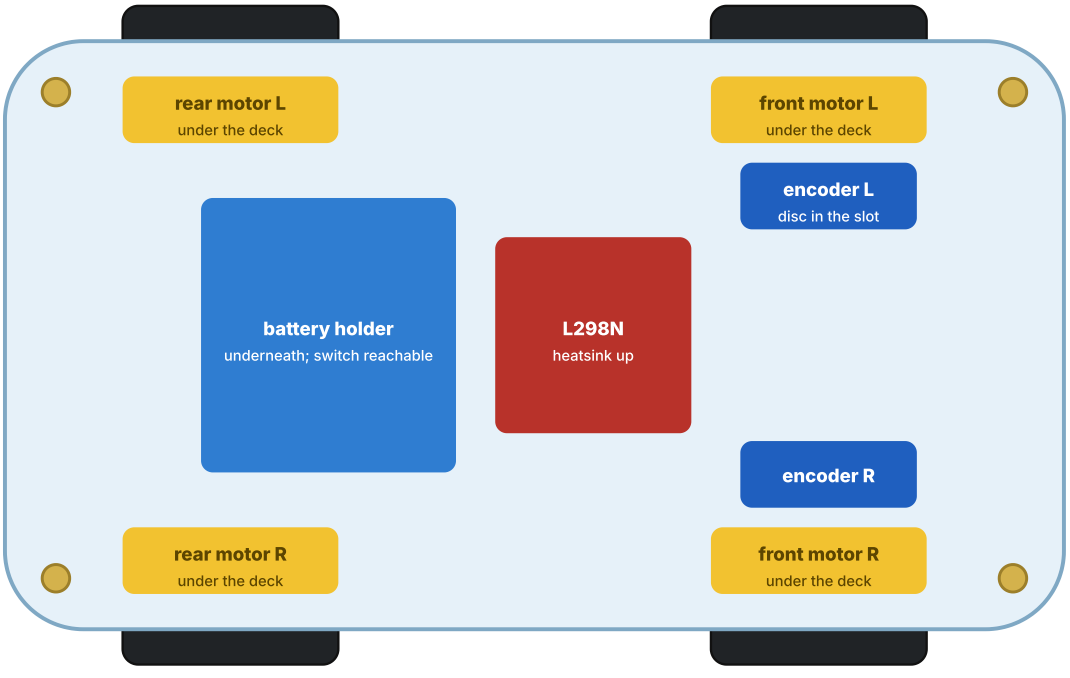


Front view: three distance sensors fanned out, the webcam above.



Bottom deck: battery holder, L298N, the 4 motors and encoders. Keep AHT20 and MQ-135 away from the Pi CPU.

Top deck.

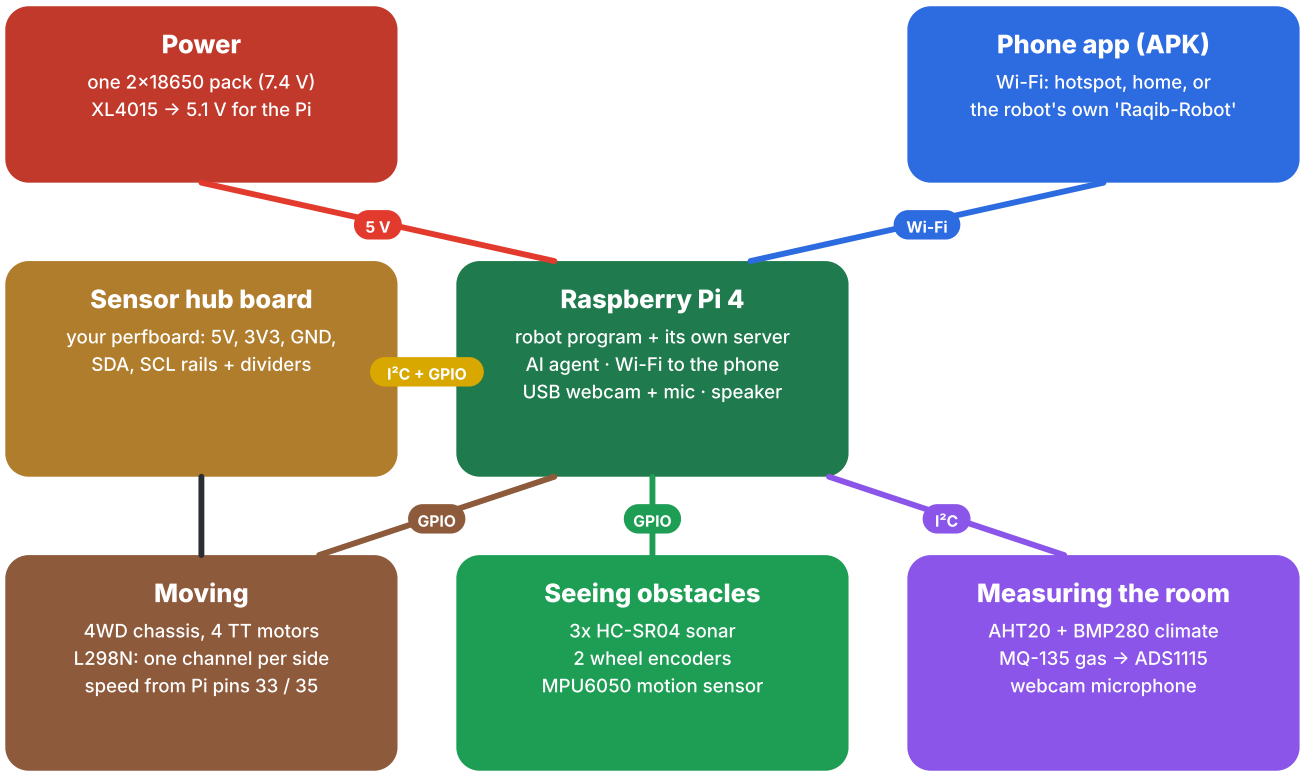


Gold dots = standoffs to the top deck. Motor and battery wires go up through the deck holes.

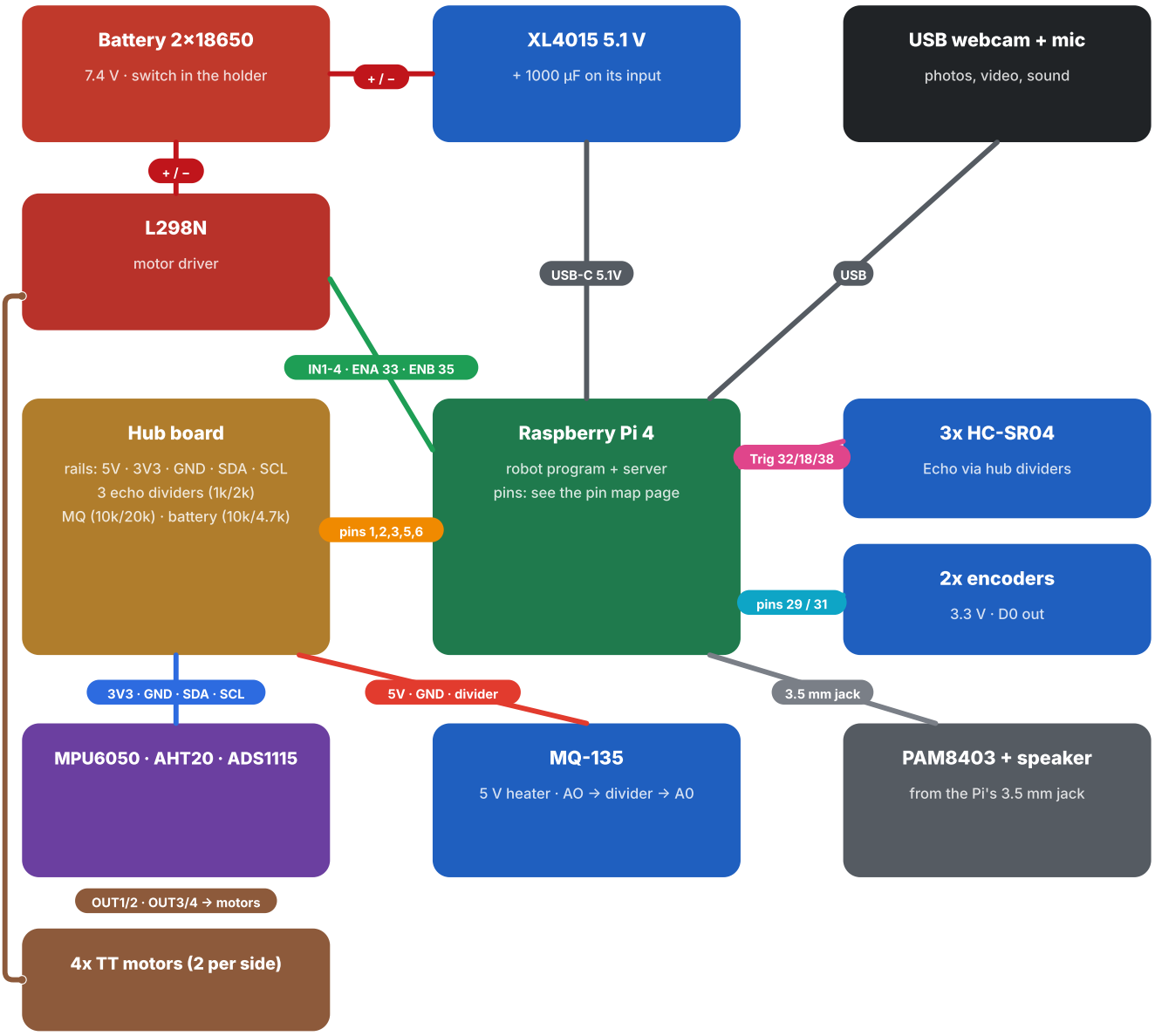
Bottom deck.

D How everything connects

Start with the big picture, then one diagram per group. The pin numbers are the physical pins of the Raspberry Pi header.



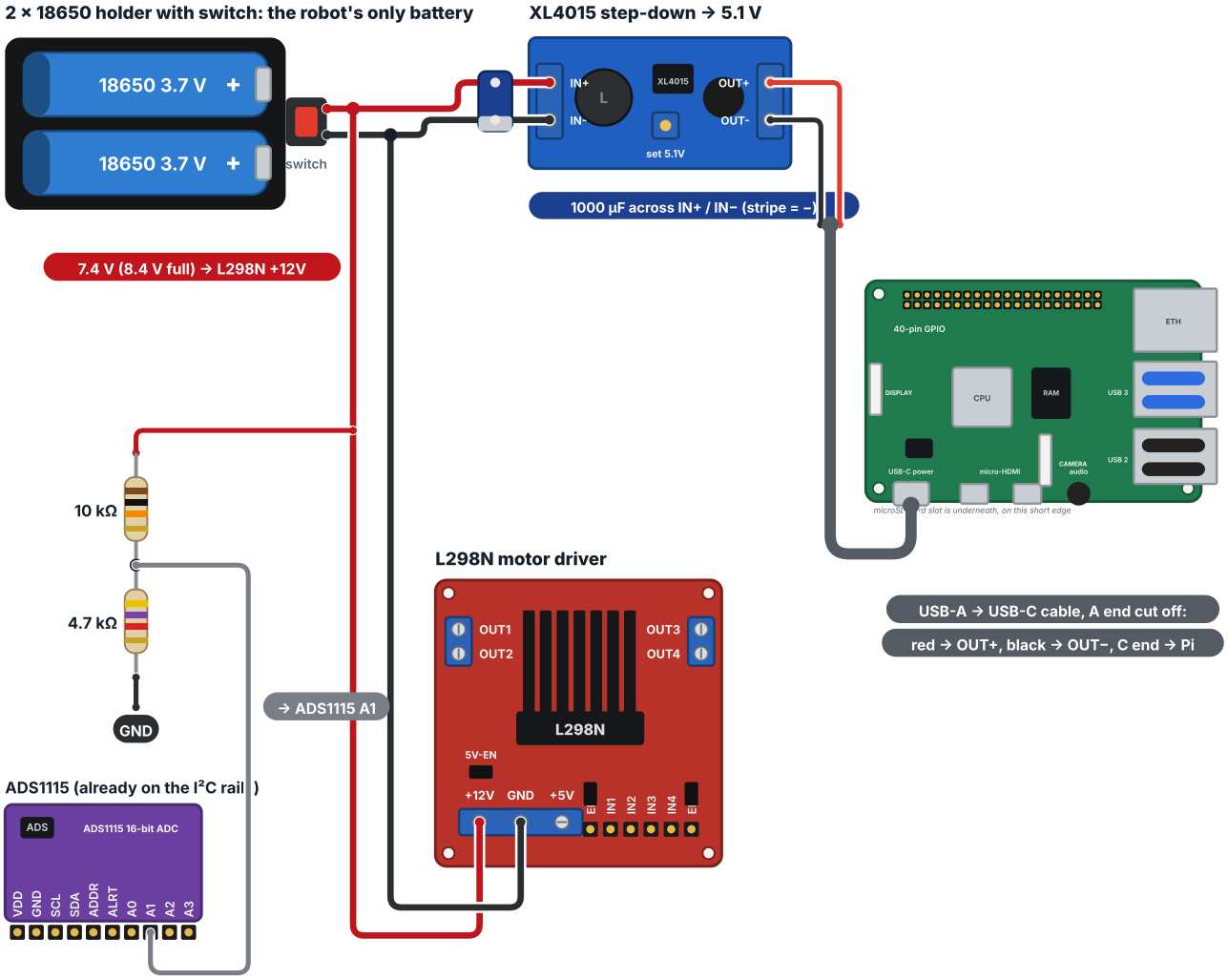
The system: five groups of parts around the Raspberry Pi.



Echo lines: sonar Echo → hub divider → Pi pins 36 / 22 / 40. Battery + → 10k/4.7k divider → ADS1115 A1.

The complete harness: every block and the wires between them.

1 · Power

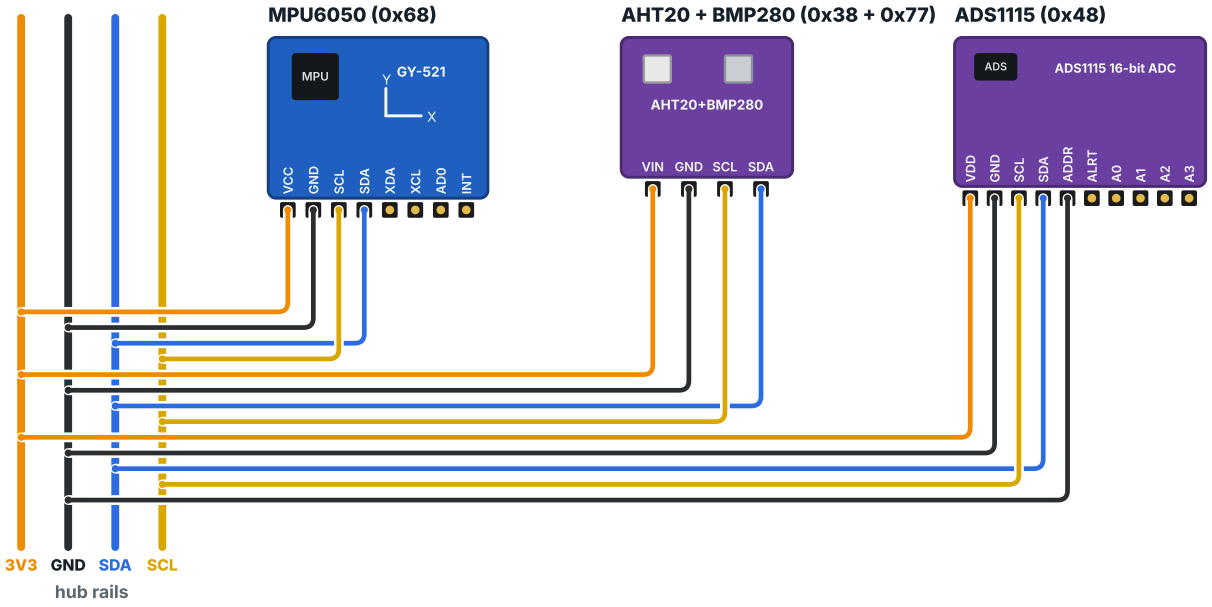


One pack feeds everything: the L298N takes the battery straight, the XL4015 makes 5.1 V for the Pi. The 1000 µF capacitor stops motor starts from dipping the Pi's supply. All grounds are joined through the battery − wire.

Set the XL4015 to 5.1 V with the multimeter BEFORE the Pi is plugged in. The divider lets the app show the battery %.

Battery → XL4015 → Pi; battery → L298N; battery divider → ADS1115 A1.

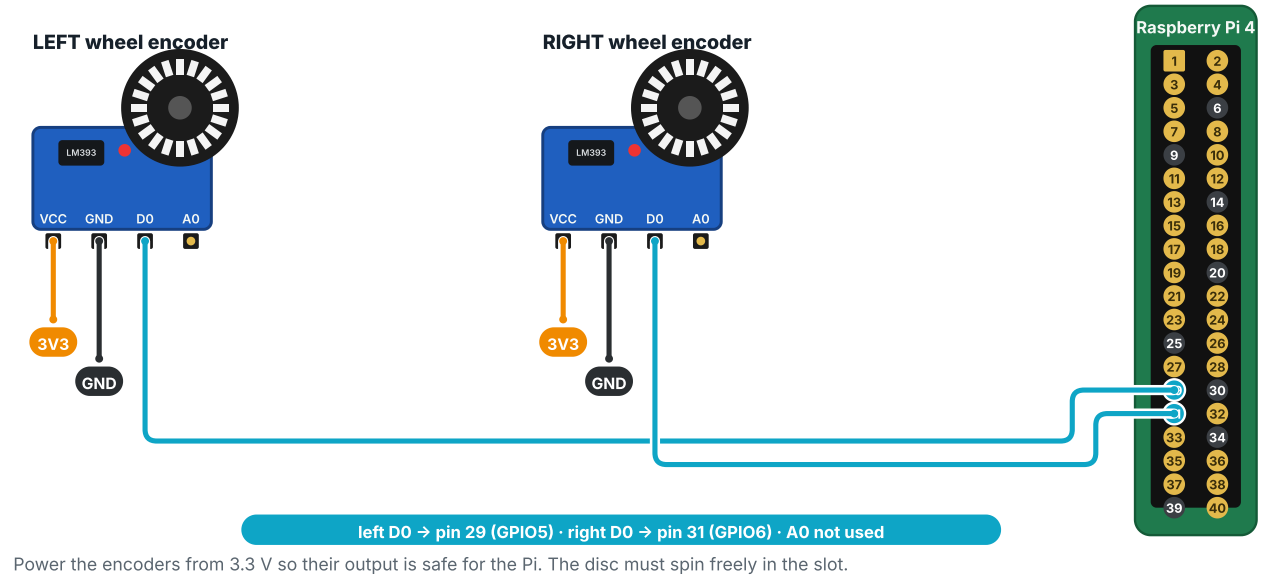
2 · I²C sensors



- Every I²C board joins the same four rails: 3.3 V, GND, SDA, SCL.
- Each answers to its own address, so they share two wires without clashing.
- ADS1115 ADDR → GND sets 0x48. A0 = gas sensor, A1 = battery voltage.
- The AHT20 + BMP280 combo board shows as two addresses: 0x38 and 0x77 (or 0x76).
- Test: python -m hwtest.i2c_scan lists every address found.

MPU6050, AHT20 + BMP280 and ADS1115 share four wires.

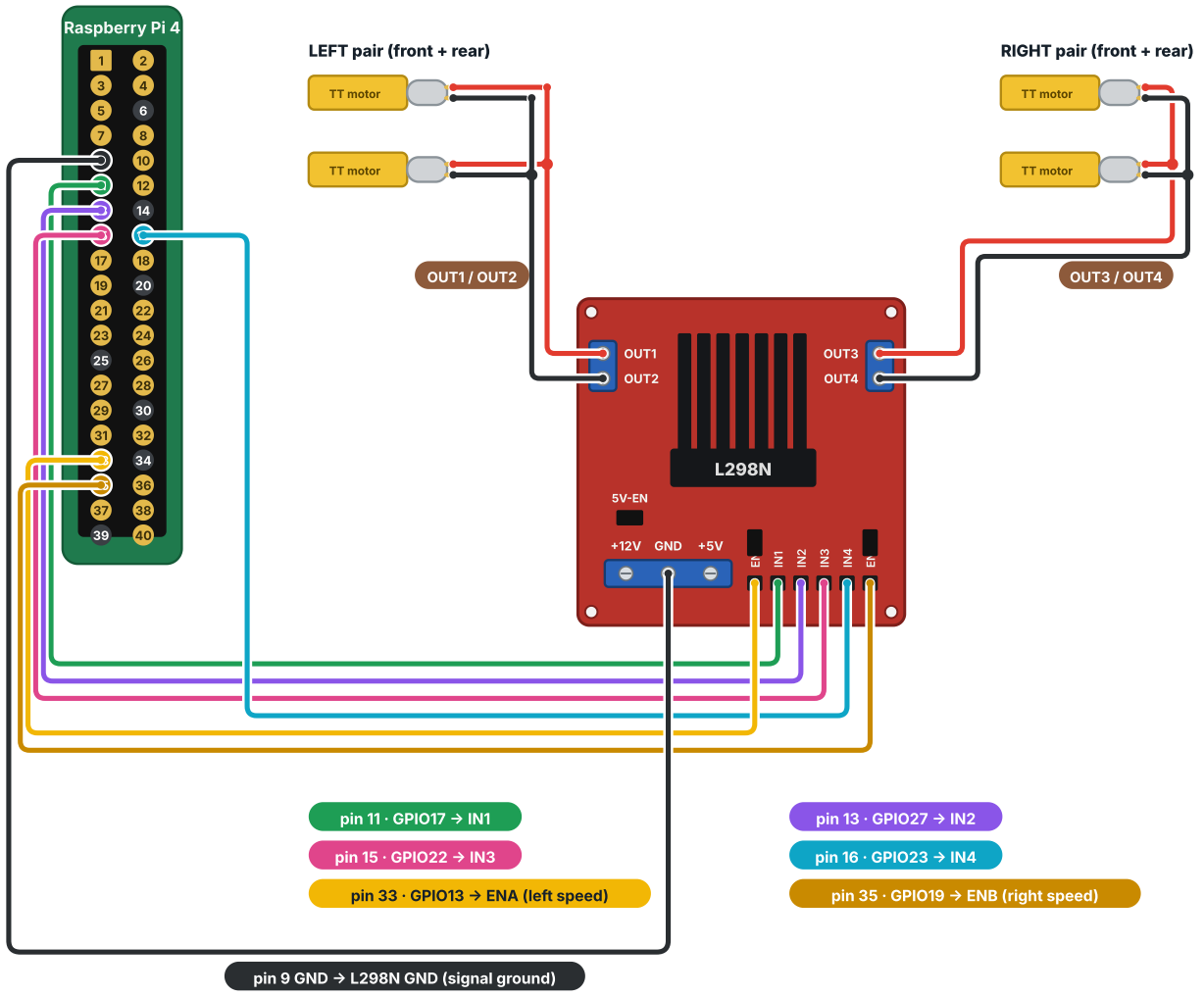
3 · Wheel encoders



Power the encoders from 3.3 V so their output is safe for the Pi. The disc must spin freely in the slot.

Encoders on pins 29 and 31.

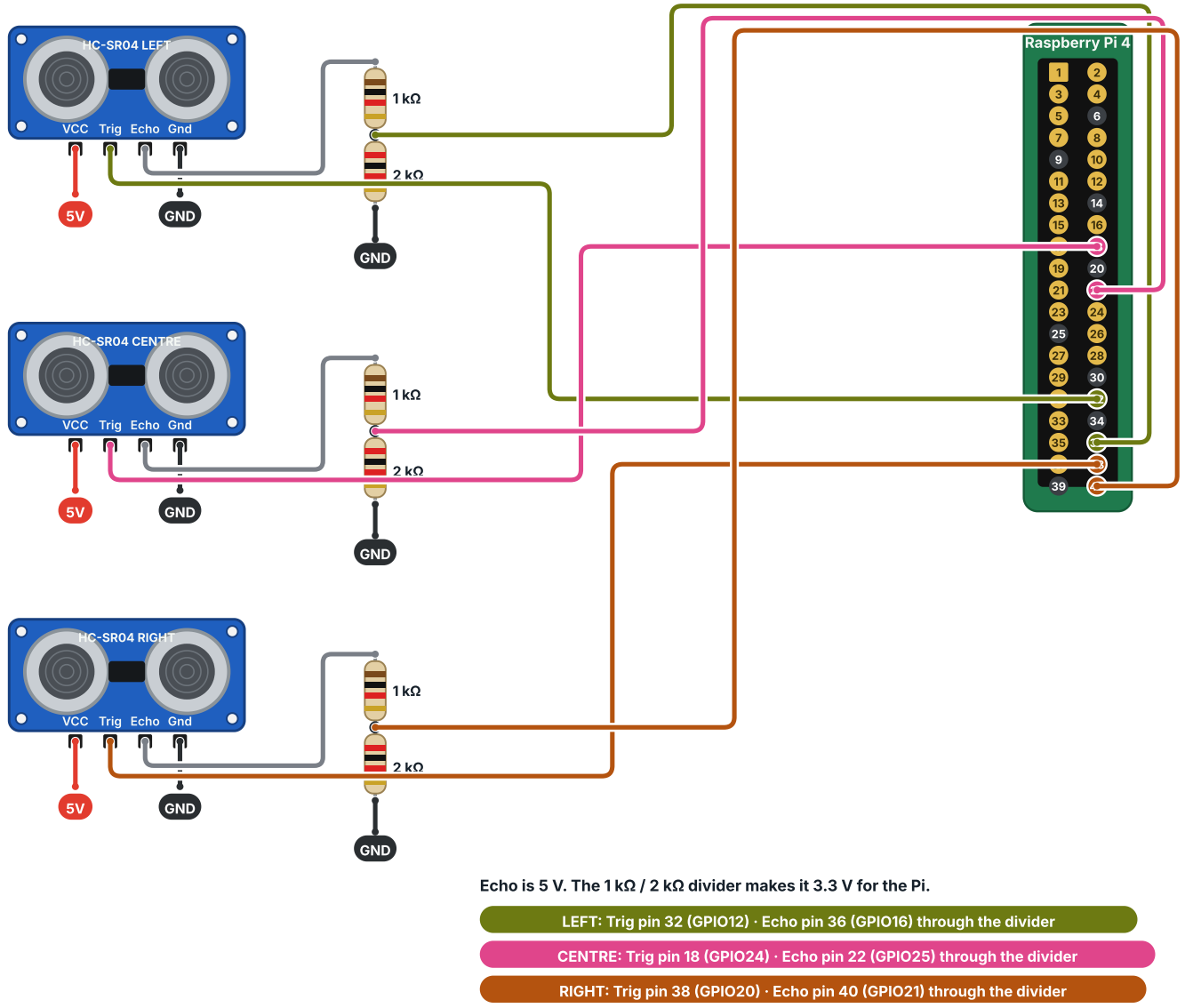
4 · Motors



- Pull the ENA and ENB jumpers off: the Pi sets the speed on pins 33 and 35. Keep the 5V-EN jumper.
- +12V and GND come from the battery (power page). Leave the L298N's +5V terminal unconnected.
- Each side's two motors are joined (red to red, black to black) and run together: skid steering.

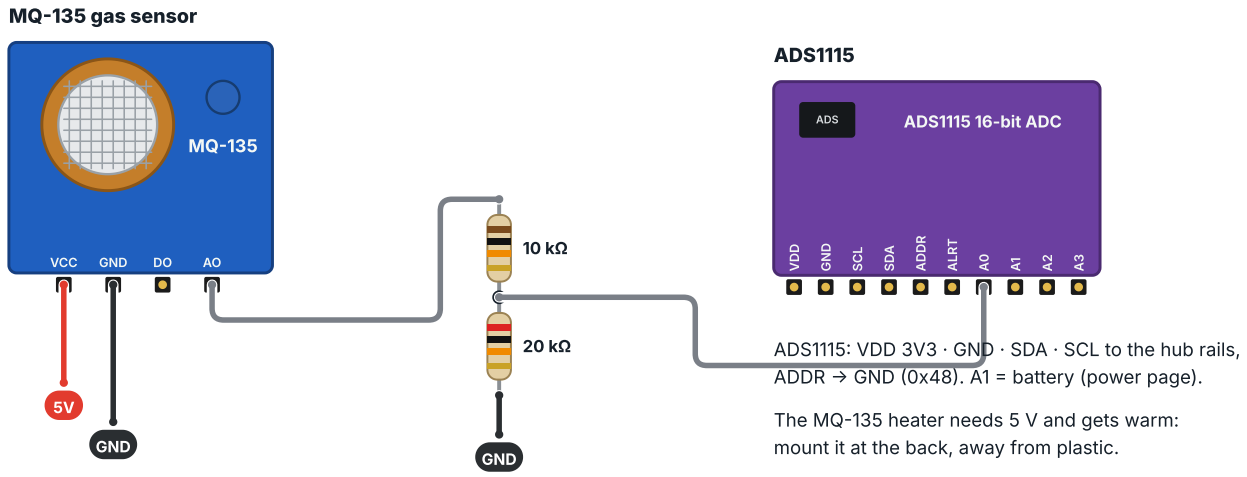
Direction on pins 11/13/15/16, speed on pins 33/35.

5 · Distance sensors



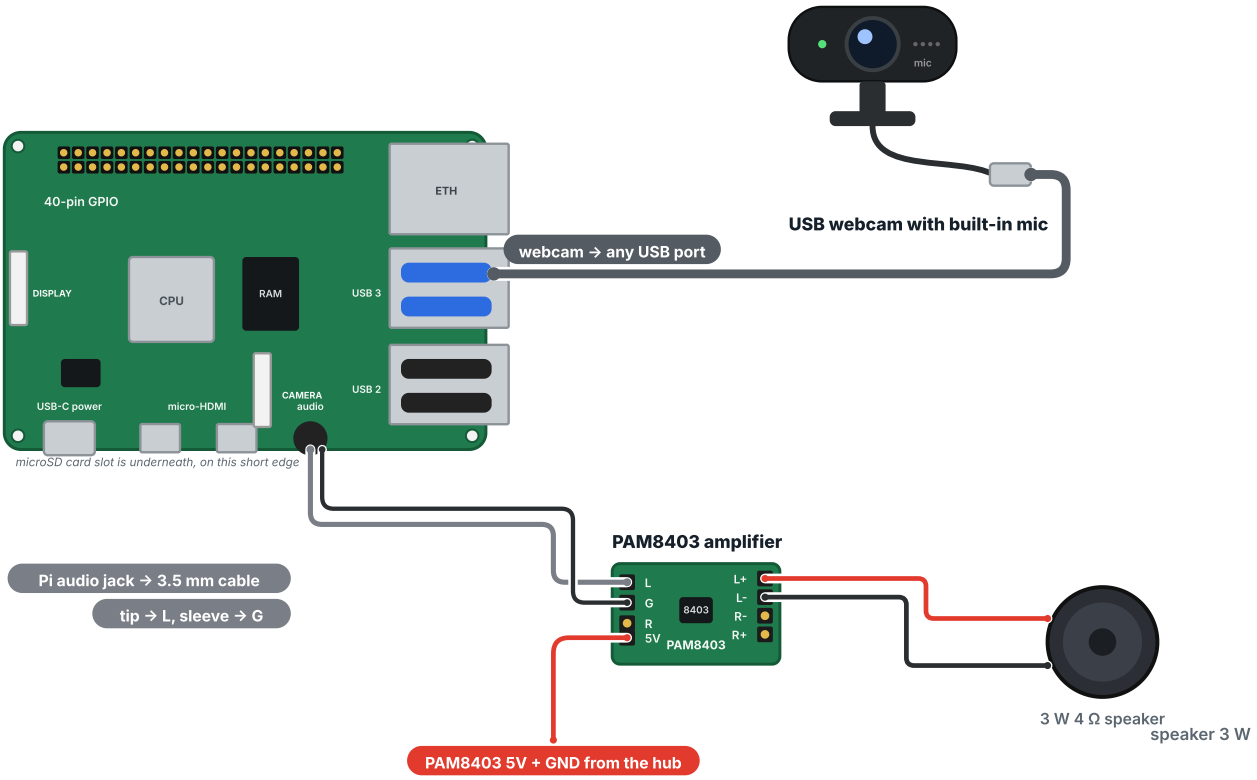
Each Echo goes through a 1 kΩ / 2 kΩ divider.

6 · Gas sensor



MQ-135 → 10 kΩ / 20 kΩ divider → ADS1115 A0.

7 · Camera, microphone and speaker



Webcam on USB; Pi audio jack → PAM8403 → speaker.

Every wire, one by one

The same list the robot software and the app's Wiring Bench use. Tick each wire off as you connect it.

Part	Pin	Goes to	Wire
XL4015 5 V converter	**IN+**	Battery +	red
	IN-	Battery -	black
	OUT (cut USB cable)	USB-C power input	USB cable
L298N motor driver	**+12V**	Battery +	red
	GND (battery)	Battery -	black
	GND (to Pi)	GND (hub GND rail / any Pi GND pin)	black
	ENA	Pi pin 33 (GPIO13)	any colour
	IN1	Pi pin 11 (GPIO17)	any colour
	IN2	Pi pin 13 (GPIO27)	any colour
	ENB	Pi pin 35 (GPIO19)	any colour
	IN3	Pi pin 15 (GPIO22)	any colour
	IN4	Pi pin 16 (GPIO23)	any colour
TT gear motors (4)	**Left pair**	OUT1 + OUT2	motor wires
	Right pair	OUT3 + OUT4	motor wires
HC-SR04 ultrasonic - Left	**VCC**	5 V (hub 5V rail / Pi pin 2 or 4)	red
	GND	GND (hub GND rail / any Pi GND pin)	black
	TRIG	Pi pin 32 (GPIO12)	any colour
	ECHO	Pi pin 36 (GPIO16) · through its divider	any colour
HC-SR04 ultrasonic - Center	**VCC**	5 V (hub 5V rail / Pi pin 2 or 4)	red
	GND	GND (hub GND rail / any Pi GND pin)	black
	TRIG	Pi pin 18 (GPIO24)	any colour
	ECHO	Pi pin 22 (GPIO25) · through its divider	any colour
HC-SR04 ultrasonic - Right	**VCC**	5 V (hub 5V rail / Pi pin 2 or 4)	red
	GND	GND (hub GND rail / any Pi GND pin)	black
	TRIG	Pi pin 38 (GPIO20)	any colour
	ECHO	Pi pin 40 (GPIO21) · through its divider	any colour
Wheel encoders (2)	**VCC (both)**	3.3 V (hub 3V3 rail / Pi pin 1 or 17)	orange
	GND (both)	GND (hub GND rail / any Pi GND pin)	black
	Left OUT	Pi pin 29 (GPIO5)	any colour
	Right OUT	Pi pin 31 (GPIO6)	any colour
MPU6050 IMU	**VCC**	3.3 V (hub 3V3 rail / Pi pin 1 or 17)	orange
	GND	GND (hub GND rail / any Pi GND pin)	black
	SDA	Pi pin 3 (GPIO2 SDA) · via hub SDA rail	blue
	SCL	Pi pin 5 (GPIO3 SCL) · via hub SCL rail	yellow
AHT20 + BMP280 climate sensor	**VIN**	3.3 V (hub 3V3 rail / Pi pin 1 or 17)	orange
	GND	GND (hub GND rail / any Pi GND pin)	black
	SDA	Pi pin 3 (GPIO2 SDA) · via hub SDA rail	blue
	SCL	Pi pin 5 (GPIO3 SCL) · via hub SCL rail	yellow

Part	Pin	Goes to	Wire
ADS1115 ADC	**VDD**	3.3 V (hub 3V3 rail / Pi pin 1 or 17)	orange
	GND	GND (hub GND rail / any Pi GND pin)	black
	SDA	Pi pin 3 (GPIO2 SDA) · via hub SDA rail	blue
	SCL	Pi pin 5 (GPIO3 SCL) · via hub SCL rail	yellow
	ADDR	GND (hub GND rail / any Pi GND pin)	black
MQ-135 gas sensor	**VCC (heater)**	5 V (hub 5V rail / Pi pin 2 or 4)	red
	GND	GND (hub GND rail / any Pi GND pin)	black
	AOUT	A0 (analog in) · through its divider	via divider
Battery divider (10k / 4.7k)	**Top (10 kOhm)**	Battery +	red
	Bottom (4.7 kOhm)	GND (hub GND rail / any Pi GND pin)	black
	Middle point	A1 (analog in) · through its divider	via divider
USB webcam with microphone	**USB plug**	USB port	USB cable
Speaker + PAM8403 amp	**Amp VCC**	5 V (hub 5V rail / Pi pin 2 or 4)	red
	Amp GND	GND (hub GND rail / any Pi GND pin)	black
	Audio in (L)	3.5 mm audio jack	audio cable